

2. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with 0 and end with 1

Code:

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    printf("Enter binary string: ");
    scanf("%s", s);

    int n = strlen(s);

    if (n > 0 && s[0] == '0' && s[n - 1] == '1')
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```

Output:

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    printf("Enter binary string: ");
    scanf("%s", s);

    int n = strlen(s);

    if (n > 0 && s[0] == '0' && s[n - 1] == '1')
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```

Enter binary string: 0101
Accepted

=== Code Execution Successful ===

